King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam I Term 241

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Which one of the following statements is TRUE about $f(x) = \dfrac{x^3 + x}{x^2 + x}$?

(a) f has a removable discontinuity at $x = 1$.

(b) f has a nonremovable discontinuity at $x = -1$.

(c) f has a nonremovable discontinuity at $x = 0$.

(d) f has a removable discontinuity at $x = -1$.

(e) f has a nonremovable discontinuity at $x = 1$.

2. If the line $y = 3x + 1$ is tangent to the graph of $f(x) = kx^2$, then

(a) $k = \dfrac{3}{2}$

(b) $k = -\dfrac{9}{4}$

(c) $k = -\dfrac{3}{2}$

(d) $k = \dfrac{9}{2}$

(e) $k = \dfrac{5}{4}$

3. $\lim_{x \to 5^-} \dfrac{x}{x - 5} =$

(a) ∞

(b) $\dfrac{1}{5}$

(c) 1

(d) 5

(e) $-\infty$

4. If $g(x) = \begin{cases} \sqrt{1-x}, & x \leq 1 \\ x+1, & x > 1 \end{cases}$ then $\lim_{x\to 1} g(x)$

(a) is equal to 0

(b) is equal to 2

(c) is equal to $\frac{1}{2}$

(d) is equal to 1

(e) does not exist

5. $\lim_{x\to 4} \frac{x - 2\sqrt{x}}{x^2 - 3x - 4} =$

(a) $\frac{1}{20}$

(b) $\frac{1}{5}$

(c) $\frac{1}{10}$

(d) $\frac{4}{5}$

(e) $\frac{5}{8}$

6. The graph of $f(x) = \frac{x^4}{x^3 - 9x}$ has (V.A: vertical asymptote(s); H.A: horizontal asymptote(s))

(a) 3 V. A. and 1 H. A.

(b) 2 V. A. and no H. A.

(c) 2 V. A. and 1 H. A.

(d) 3 V. A. and no H. A.

(e) no V. A. and 1 H. A.

7. Which one of the following statements is TRUE? (c and L are real numbers)

(a) If $\lim_{x\to c} f(x) = L$, then $f(c) = L$.

(b) If $f(c)$ is undefined, then $\lim_{x\to c} f(x)$ does not exist.

(c) If $\lim_{x\to c} f(x) = f(c)$, then f is continuous at c.

(d) If $\lim_{x\to c} f(x) = \infty$, then $f(c)$ is undefined.

(e) If $f(c) = L$, then $\lim_{x\to c} f(x) = L$.

8. $\lim_{x\to-\infty}\left(5e^x-3x^3\right)=$

(a) 0

(b) 2

(c) $-\infty$

(d) -2

(e) ∞

9. If $y=Ax+B$ is an equation of the tangent line to the graph of $f(x)=8\sqrt{x}+\dfrac{3}{x^2}$ at the point $(1,11)$, then $A-B=$

(a) -9

(b) 0

(c) -11

(d) 4

(e) -15

10. Find the interval(s) on which $f(x)=\dfrac{\sqrt{x+3}}{x^2+4x}$ is continuous?

(a) $[-3,0)\cup(0,\infty)$

(b) $(-\infty,-4)\cup(0,\infty)$

(c) $(-4,-3]$

(d) $(-4,-3)\cup(-3,0)\cup(0,\infty)$

(e) $(-\infty,-4)\cup[-3,\infty)$

11. $\lim_{x\to\frac{1}{2}}\left(\lfloor 2x\rfloor+\lfloor -2x\rfloor\right)$

(a) is equal to 2

(b) is equal to -1

(c) does not exist

(d) is equal to 0

(e) is equal to 1

12. $\lim_{x\to 0} \dfrac{e^{3x}-1}{e^x-1} =$

(a) 3

(b) ∞

(c) 0

(d) 2

(e) 1

13. $\lim_{x\to \infty} \dfrac{\sqrt{x^4-1}}{x^3-1} =$

(a) -1

(b) ∞

(c) $-\infty$

(d) 0

(e) 1

14. If the function $f(x) = \begin{cases} \dfrac{\tan(3x)}{x}, & x < 0 \\ 6a - x^2, & x \geq 0 \end{cases}$ is continuous at $x = 0$, then $a =$

(a) $\dfrac{1}{2}$

(b) $\dfrac{1}{3}$

(c) 2

(d) -1

(e) 0

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$x = 0$ removable (factor cancels); $x = -1$ nonremovable (infinite limit).
2	**(b)**	$2kx_0 = 3$ and $kx_0^2 = 3x_0 + 1$ give $x_0 = -\frac{2}{3}$, $k = -\frac{9}{4}$.
3	**(e)**	Numerator $\to 5$, denominator $\to 0^-$: limit $-\infty$.
4	**(e)**	Left limit 0, right limit 2, so the limit does not exist.
5	**(c)**	Conjugate and factor: $\frac{x}{(x+1)(x+2\sqrt{x})} \to \frac{4}{40} = \frac{1}{10}$.
6	**(b)**	V.A. at $x = \pm 3$ ($x = 0$ is a hole); degree $3 > 2$, so no H.A.
7	**(c)**	Continuity is defined by $\lim_{x\to c} f(x) = f(c)$; the others fail by holes or displaced points.
8	**(e)**	$5e^x \to 0$, $-3x^3 \to \infty$: limit ∞.
9	**(e)**	$A = f'(1) = -2$, $B = 13$, so $A - B = -15$.
10	**(a)**	Need $x \geq -3$ and $x \neq 0, -4$: $[-3, 0) \cup (0, \infty)$.
11	**(b)**	Left: $0 + (-1) = -1$; right: $1 + (-2) = -1$; limit -1.
12	**(a)**	$\frac{(e^x-1)(e^{2x}+e^x+1)}{e^x-1} \to 1 + 1 + 1 = 3$.
13	**(d)**	Degree 2 over degree 3: limit 0.
14	**(a)**	Left limit 3, $f(0) = 6a$: $a = \frac{1}{2}$.